

GPH NCC PORTAL

A Report Submitted in Partial Fulfillment

Of the Requirement For the Award Of

DIPLOMA IN INFORMATION TECHNOLOGY

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JANUARY 2026 -MAY 2026

Certificate

We hereby certify that the work which is being presented in the Diploma Major Project Report entitled “**National Cadet Corps Portal**”, in partial fulfillment of the requirements for the award of the Diploma in Information Technology, and submitted to the Department of Information Technology, Government Polytechnic Hamirpur, is an authentic record of work carried out during the period Feb 2026 to June 2026 under the supervision of Mr. Vijay Pathania and Mr.Madhusudhan Lecturer Information Technology.

The material contained in this report has not been submitted by us for the award of any other diploma elsewhere.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported and guided me throughout the development of this Library Management System project.

First and foremost, I would like to thank Mr. Vijay Pathania and Mr. Madhusudhan for their valuable guidance, continuous encouragement, and insightful feedback that greatly contributed to the successful completion of this project.

I am also thankful to Govt. Polytechnic Hamirpur for providing the necessary facilities and a conducive environment for carrying out this project work.

My heartfelt thanks also go to my friends and classmates for their suggestions and moral support during the various stages of the project.

This project has helped me deepen my understanding of software development and database management, and I am grateful for the opportunity to apply theoretical knowledge to a practical application.

Abstract

The National Cadet Corps Website is a modern web-based platform developed for managing the activities and administration of the National Cadet Corps (NCC) unit at Govt. Polytechnic Hamirpur (HP). The project digitizes the traditional manual management system and provides an efficient online platform for cadet enrolment, notices, camp management, achievements, gallery management, and communication.

The system is developed using Flask for the backend and HTML, CSS, and JavaScript for the frontend. The website improves accessibility, transparency, and management efficiency for students, cadets, and NCC administrators.

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List of Abbreviations

S.No.	Abbreviation	Full Form
1	NCC	National Cadet Corps
2	HTML	HypertextMarkup Language
3	CSS	Cascading Style Sheets
4	JS	JavaScript
5	PWA	Progressive Web Application
6	PDF	Portable Document Format
7	QR	Quick Response
8	API	Application Programming Interface
9	DB	Database
10	SQL	Structured Query Language
11	SQLite	Structured Query Language Lite
12	RAM	Random Access Memory
13	SSD	Solid State Drive
14	GB	Gigabyte
15	VS Code	Visual Studio Code
16	HP	Himachal Pradesh
17	IT	Information Technology
18	SMS	Short Message Service
19	AI	Artificial Intelligence

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1. Introduction

The NCC Website System is a complete web-based application developed for managing the activities and records of National Cadet Corps (NCC) cadets in an educational institution. Traditionally, most NCC-related activities such as cadet registration, certificate handling, attendance management, notices, and report generation are maintained manually using paper records or spreadsheets. Manual systems are time-consuming, difficult to manage, and prone to data loss and human errors.

The proposed NCC Website System digitalizes the complete workflow of NCC administration and provides an efficient platform for cadets and administrators. The project is developed using modern web technologies including HTML, CSS, JavaScript, Python Flask framework, SQLite database, PDF generation utilities, QR code integration, and Excel report generation.

The system provides different modules such as cadet registration, login authentication, certificate management, admin dashboard, QR code verification, PDF generation, and Excel export facilities. It also includes a responsive user interface that improves usability on both desktop and mobile devices.

The project aims to reduce paperwork, improve accuracy, maintain secure records, and simplify NCC management activities. The website allows administrators to manage cadet data efficiently while cadets can easily access notices, certificates, and registration services.

The implementation of this system demonstrates practical usage of full-stack web development technologies and database management techniques. The project is scalable and can be extended further with cloud integration, biometric attendance, mobile applications, and advanced analytics.

2. Overview of NCC Website System

2.1 Introduction to NCC

The National Cadet Corps (NCC) is the youth wing of the Indian Armed Forces. It is responsible for developing discipline, leadership, patriotism, and character among students. NCC organizes training programs, camps, parades, social service activities, adventure programs, and national-level events.

Managing NCC activities manually becomes difficult when the number of cadets increases. Important records such as enrollment details, camp participation, certificates, and notices require proper maintenance. The NCC Website System helps automate these operations and provides a centralized platform for administrators and cadets.

The website acts as an information management and communication platform where cadets can register, access updates, and manage their information. Administrators can maintain cadet databases, verify records, generate certificates, and export reports.

2.2 Objectives of the Project

The main objectives of the NCC Website System are:

- To digitalize NCC administrative operations.
- To reduce manual paperwork and data redundancy.
- To provide an easy registration system for cadets.
- To manage certificates digitally.
- To generate QR-code-based verification systems.
- To provide PDF and Excel export functionality.
- To create a responsive and user-friendly interface.
- To improve security and data accuracy.
- To simplify communication between cadets and administrators.
- To maintain centralized storage of NCC records.

2.3 Features of the Website

The project contains several useful features that improve the efficiency of NCC management.

Main Features:

1. Responsive Homepage
2. Cadet Registration and Login
3. Admin Dashboard
4. Certificate Management System
5. QR Code Generation and Verification
6. PDF Report Generation
7. Excel Export Functionality
8. Secure Authentication System
9. Notice Management
10. Activity and Gallery Sections
11. Contact and About Pages
12. Database Backup Support
13. Mobile-Friendly Interface
14. Dynamic Data Handling
15. Upload and Storage Facilities

The website also includes service worker and manifest files, which indicate support for Progressive Web Application (PWA) functionality.

2.4 Scope of the Project

The scope of the project is broad and highly useful for educational institutions that operate NCC units.

Scope Includes:

- Automation of NCC administrative work.
- Centralized management of cadet information.
- Digital certificate verification.
- Reduction in paperwork.
- Better communication system.
- Report generation and analytics.
- Secure storage of student records.
- Future scalability for larger institutions.

The project can be extended for state-level or national-level NCC management systems in the future.

3. System Requirements and Technologies Used

3.1 Hardware Requirements

The project requires standard hardware components for development and deployment.

Minimum Hardware Requirements:

Component	Specification
Processor	Intel i3 or higher
RAM	4 GB minimum
Storage	20 GB free space
Internet	Required for deployment
Display	1366 × 768 resolution

Recommended Hardware Requirements:

Component	Specification
Processor	Intel i5/i7
RAM	8 GB or higher
Storage	SSD recommended
Internet	High-speed connection

3.2 Software Requirements

The project uses various software tools and libraries.

Software	Purpose
Python	Backend programming
Flask	Web framework
SQLite	Database management
HTML5	Structure of website
CSS3	Styling and responsiveness
JavaScript	Interactive frontend

VS Code Development environment

Git Version control

Browser Testing and execution

The backend also uses additional utilities for PDF generation, QR code generation, and Excel export functionality.

3.3 Frontend Technologies

The frontend of the project is designed using modern web technologies.

HTML5

HTML5 is used for creating the structure of web pages. It defines forms, sections, navigation bars, images, buttons, and content areas.

CSS3

CSS3 is used to design the website and improve visual appearance. The project includes a dedicated style.css file containing responsive layouts, animations, colors, spacing, typography, and grid structures.

JavaScript

JavaScript provides interactivity and dynamic functionality. It is used for form validation, animations, event handling, and frontend logic.

Responsive Design

The website supports responsive design for mobile and desktop users.

PWA Support

The presence of manifest.json and sw.js files indicates Progressive Web Application support.

3.4 Backend Technologies

Python

Python is the primary backend programming language used in the project. It is easy to maintain, secure, and highly scalable.

Flask Framework

Flask is a lightweight Python web framework used to create routes, APIs, authentication systems, and backend logic.

Modules Used in Backend

The backend contains multiple modules such as:

- app.py
- models.py
- email_utils.py
- excel_utils.py
- pdf_utils.py
- qr_utils.py
- config.py
- db_backup.py

These modules improve modularity and maintainability.

API and Routing

Flask routes manage user requests and responses. Backend APIs interact with the frontend and database.

3.5 Database Used

The project uses SQLite database.

Advantages of SQLite:

- Lightweight database
- No separate server required
- Easy integration with Flask
- Fast for small and medium applications
- Portable database file

The database stores:

- Cadet details
- Login credentials
- Certificate records
- Notices
- Activity data
- Admin information

The database file used in the project is:

ncc_database.db

4. System Design and Architecture

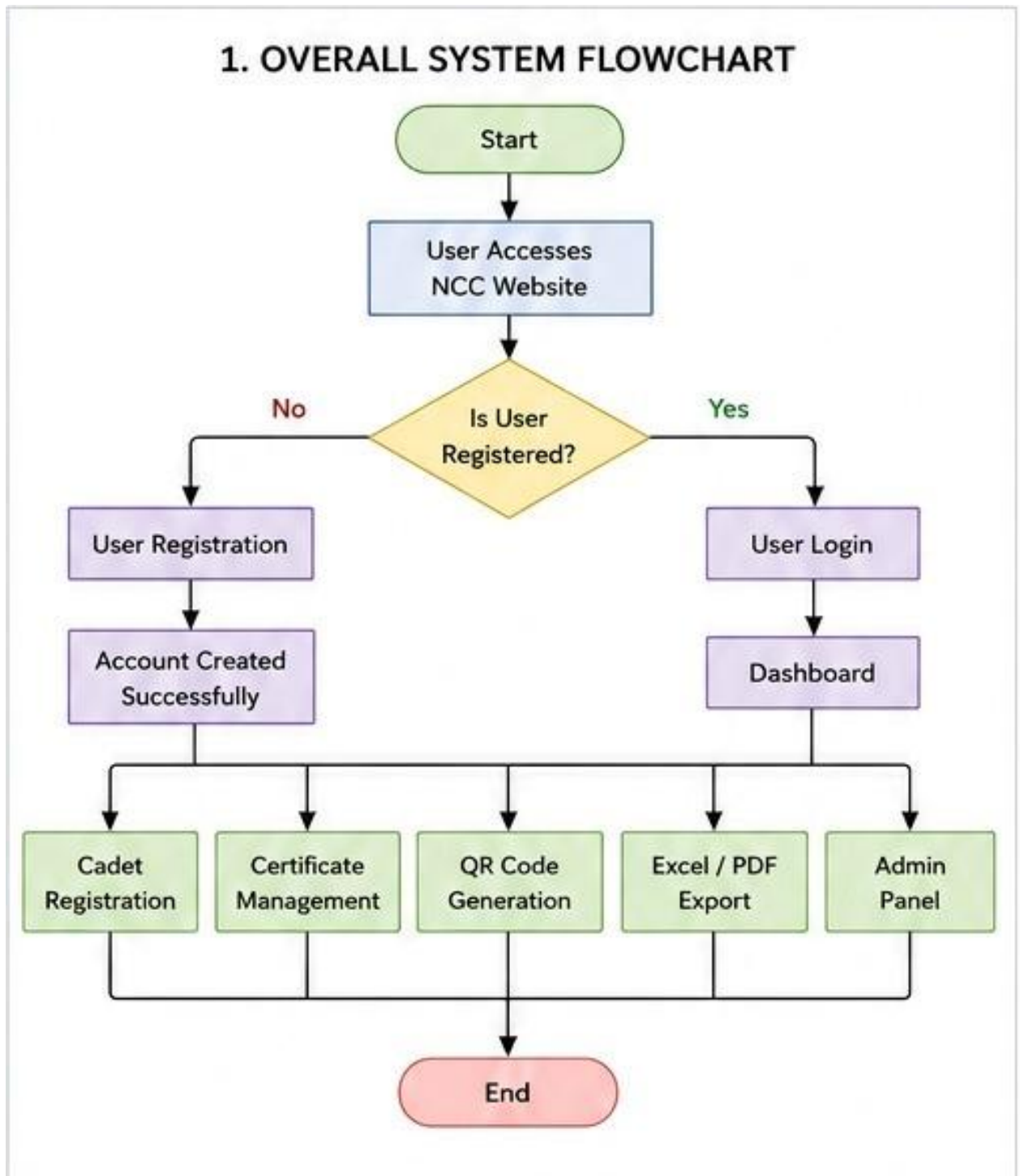


FIG : 4.1.1

4.1 Website Architecture

The NCC Website System follows a layered architecture.

Layers of Architecture:

1. Presentation Layer (Frontend)
2. Application Layer (Backend Logic)
3. Database Layer (SQLite)

Working Architecture:

User → Frontend → Flask Server → Database → Response → User

The frontend sends requests to the Flask backend. The backend processes data and communicates with the database. The database returns records which are displayed on the frontend.

This architecture ensures modularity, maintainability, and scalability.

4.2 Database Design

The database design is structured to maintain different types of records efficiently.

Main Tables:

- Cadet Table
- Admin Table
- Certificate Table
- Activity Table
- Notice Table

Database Operations:

- Insert Records
- Update Records
- Delete Records
- Search Records
- Fetch Data
- Export Data

The project uses models.py to define and manage database models.

4.3 Modules of the System

The project is divided into several modules.

1. Homepage Module

Provides navigation and information about NCC.

2. Registration Module

Allows cadets to register and store details.

3. Authentication Module

Handles login and security.

4. Certificate Module

Manages NCC certificates digitally.

5. QR Code Module

Generates QR codes for verification.

6. Admin Module

Allows administrators to manage records.

7. Report Module

Generates Excel and PDF reports.

8. Gallery and Activity Module

Displays NCC activities and achievements.

4.4 User Flow of the Website

Cadet Flow:

1. Open Website
2. Register/Login
3. Access Dashboard
4. View Notices and Activities
5. Download Certificates
6. Update Profile

Admin Flow:

1. Login to Admin Panel
2. Manage Cadet Records
3. Upload Certificates
4. Generate QR Codes
5. Export Reports
6. Monitor Activities

The flow ensures easy navigation and efficient management.

5. Implementation of the Project

5.1 Homepage Design

The homepage is designed to provide a modern and attractive interface. It includes:

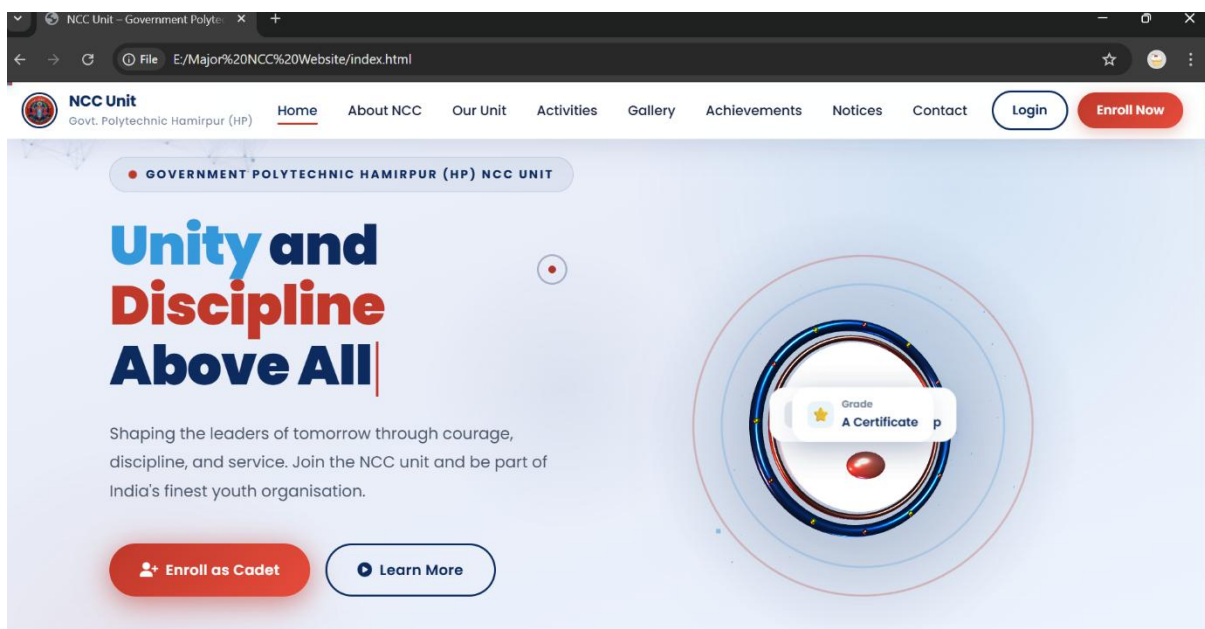
- Navigation bar
- Hero section
- NCC introduction
- Activity highlights
- Image galleries
- Notices section
- Contact information

The project includes multiple images and styling files for enhanced user experience.

Frontend Files Used:

- index.html
- style.css
- main.js

The homepage design focuses on simplicity, responsiveness, and accessibility.



5.2 Cadet Registration Module

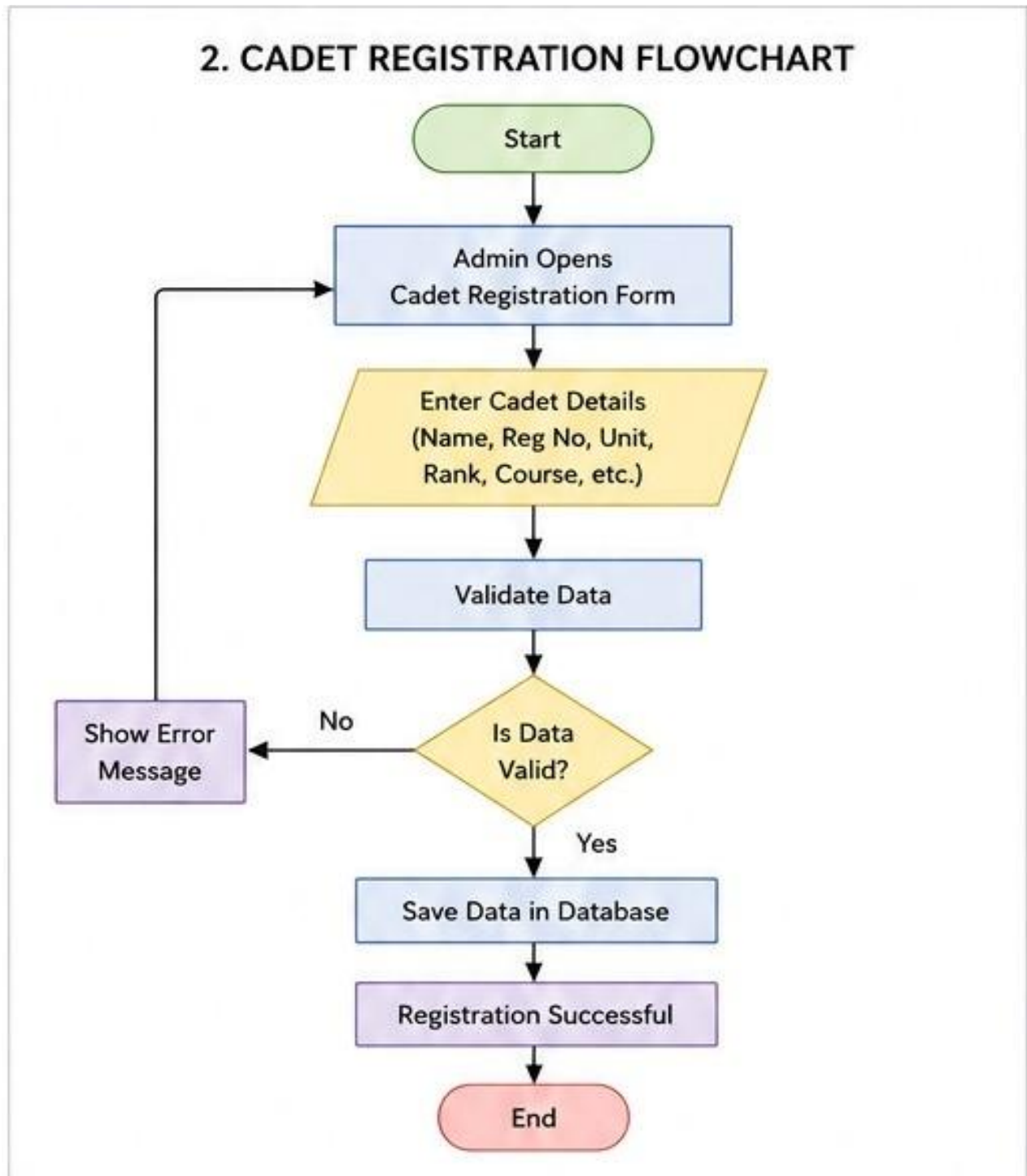


FIG : 5.2.1

The registration module allows cadets to create accounts and submit details.

Features:

- User signup
- Login system
- Validation checks
- Database storage
- Secure authentication

Information Collected:

- Name
- Enrollment Number
- Email
- Contact Number
- NCC Unit Details
- Course Information

The backend validates and stores all information securely.

The screenshot displays the NCC Unit website for Govt. Polytechnic Hamirpur (HP). The header includes a navigation menu with links: Home, About NCC, Our Unit, Activities, Gallery, Achievements, Notices, Contact, Login, and an Enroll Now button. The main banner area features the text "Join the NCC Family" and a sub-header "Take your first step towards a life of discipline, honour and service. Fill out the form to enroll as an NCC cadet."

Below the banner, there are two sections. On the left, titled "WHY JOIN NCC?", is a section "Benefits of Being a Cadet" with five bullet points: "Defence Recruitment Advantage" (C Certificate holders get relaxed physical standards in over 25 Armed Forces recruitment processes), "Stipend & Scholarships" (Cadets receive daily allowances during camps and are eligible for special government scholarships), "International Exposure" (Selected cadets participate in Youth Exchange Programmes with countries like the UK, USA, Japan and more), "Leadership Development" (Develop communication, team management and decision-making skills valued in all careers), and "Bonus Marks in Govt. Jobs" (NCC B/C certificate holders receive additional marks in many state government job recruitment tests).

On the right is the "Enrollment Form". It includes a progress bar with four steps: 1. Personal, 2. Academic, 3. NCC Details, and 4. Submit. The form fields are: First Name, Last Name, Date of Birth (dd-mm-yyyy), Gender (Select Gender), and Phone Number. A "Next Step" button is at the bottom.

5.3 Certificate Management Module

The certificate management system allows administrators to generate and manage certificates digitally.

Functions:

- Upload certificates
- Store certificate records
- Download certificates
- Verify certificates
- Maintain digital archive

This module reduces paperwork and simplifies certificate verification.

5.4 QR Code Generation System

The project contains a dedicated QR utility module named `qr_utils.py`.

Purpose of QR Codes:

- Certificate verification
- Fast data access
- Secure authentication
- Digital validation

The QR code system improves security and prevents duplication of certificates.

5.5 Excel and PDF Features

The project includes `excel_utils.py` and `pdf_utils.py` modules.

Excel Features:

- Export cadet records
- Generate reports
- Data analysis
- Attendance sheets

PDF Features:

- Certificate generation
- Printable reports

- Official documents

These features improve administrative efficiency.

5.6 Admin Panel Functions

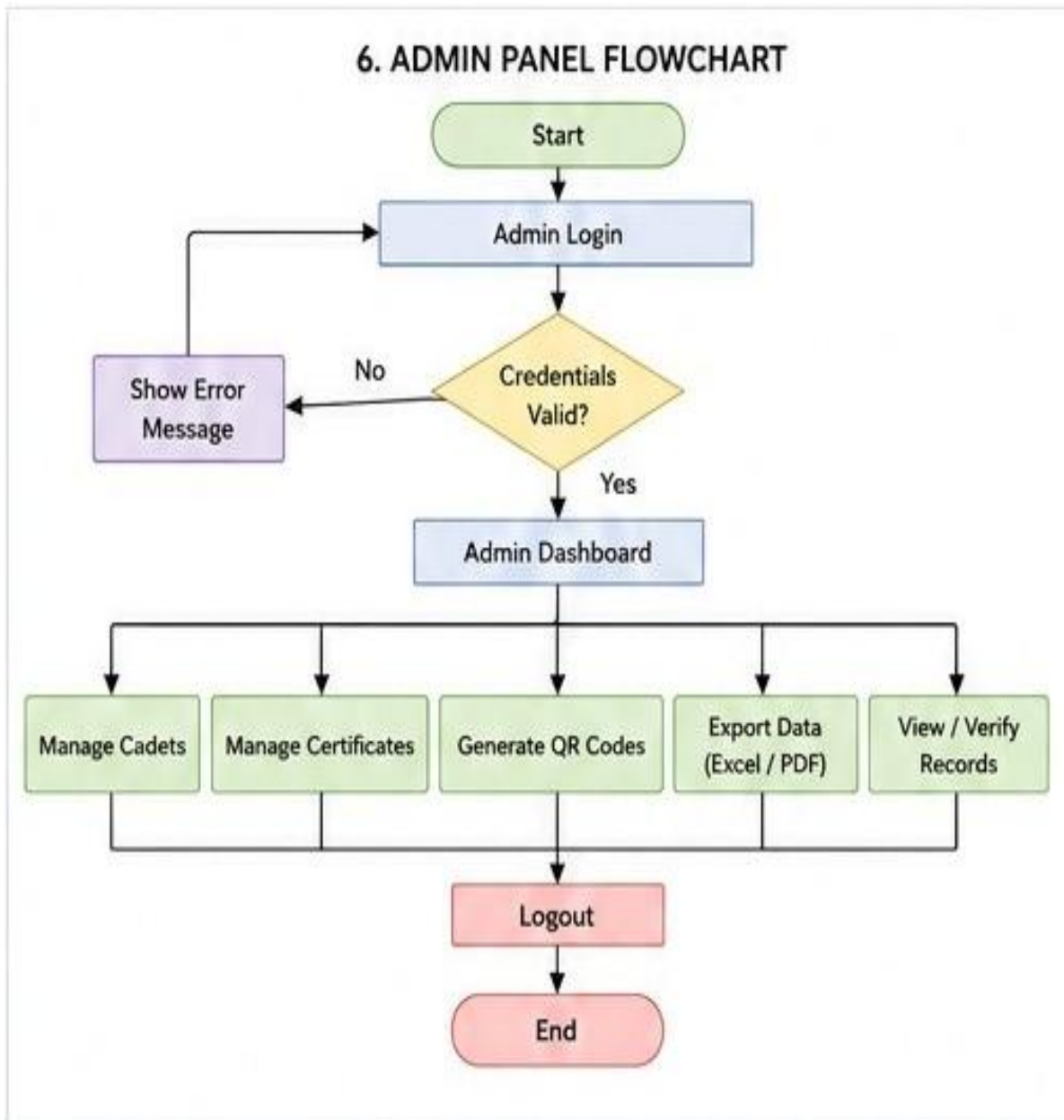


FIG : 5.6.1

The admin panel is one of the most important components of the system.

Functions of Admin Panel:

- Manage cadets
- View reports
- Upload notices
- Monitor activities
- Generate certificates
- Manage authentication
- Export data
- Backup database

The admin dashboard improves centralized management and monitoring.

6. Working of the Website

6.1 How the System Works

The website operates through interaction between frontend pages, backend Flask routes, and the SQLite database.

Working Process:

1. User opens the website.
2. Frontend displays forms and pages.
3. User submits data.
4. Backend validates the information.
5. Database stores records.
6. Backend processes requests.
7. Response is displayed to the user.

The system ensures smooth communication between all components.

6.2 Data Storage and Retrieval

The SQLite database stores all records in structured format.

Storage Process:

- User input is received.
- Backend processes form data.
- Database queries are executed.
- Records are stored securely.

Retrieval Process:

- User requests information.
- Backend fetches records.
- Results are displayed dynamically.

The project also includes backup support through db_backup.py.

6.3 Security and Authentication

Security is important in any web application.

Security Features Used:

- Login authentication
- Input validation
- Admin authorization
- Database security
- QR verification
- Controlled access

Authentication Process:

- User enters credentials.
- Backend verifies data.
- Access is granted if valid.

The project uses secure backend logic to protect records from unauthorized access.

7. Advantages and Applications

Advantages of the System

1. Reduces paperwork.
2. Improves data accuracy.
3. Provides centralized management.
4. Fast report generation.
5. Secure certificate verification.
6. User-friendly interface.
7. Saves time and effort.
8. Easy data retrieval.
9. Better communication system.
10. Scalable architecture.

Applications of the System

The project can be used in:

- Schools
- Colleges
- Universities
- NCC training centers
- Government educational institutions
- Youth organizations

It can also be adapted for other management systems such as NSS, sports clubs, and student activity portals.

8. Conclusion and Future Scope

Conclusion

The NCC Website System successfully provides a digital platform for managing NCC activities and cadet records. The project replaces traditional manual methods with an efficient and automated web-based solution.

The website integrates frontend technologies, backend programming, database management, QR code systems, PDF generation, and Excel export features into a single platform. The use of Flask framework and SQLite database makes the system lightweight, efficient, and easy to deploy.

The project demonstrates practical implementation of full-stack web development concepts and database integration. It improves efficiency, reduces paperwork, and enhances user experience.

Overall, the NCC Website System is a reliable, scalable, and user-friendly application for educational institutions.

Future Scope

The project can be further enhanced with advanced features.

Future Improvements:

1. Cloud database integration.
2. Mobile application development.
3. Biometric attendance system.
4. AI-based analytics and reports.
5. SMS and email notifications.
6. Online event registration.
7. Multi-college support.
8. Advanced dashboard analytics.
9. Real-time communication system.
10. Deployment on cloud platforms.

The system can evolve into a complete NCC management ecosystem.

9. References

Websites

1. Python Official Documentation
2. Flask Official Documentation
3. SQLite Documentation
4. MDN Web Documentation
5. W3Schools Tutorials

Technologies Referenced

- HTML5
- CSS3
- JavaScript
- Python
- Flask
- SQLite
- QR Code Libraries
- PDF Generation Libraries
- Excel Export Libraries

Development Tools

- Visual Studio Code
- GitHub
- Chrome Browser

GPH NCC PORTAL GOOGLE SITE

